Name

Mathematics Advanced Year 11

Linear Functions

Book 1

Version: 261021

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MAV-11-02 uses functions and relations to model, analyse and solve problems

Linear functions

- Determine the equations of straight lines in gradient–intercept form $y = mx + c$ with gradient m and y-intercept c
- Determine the equations of straight lines in general form $ax + by + c = 0$, where a, b and c are constants
- Determine the equation of a straight line passing through a point (x_1, y_1) with gradient m using the point–gradient formula $y - y_1 = m(x - x_1)$
- Determine the equation of a straight line passing through two points (x_1, y_1) and (x_2, y_2) by calculating its gradient m using the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$
- Determine the x-intercept and y-intercept of a straight line given its equation
- Choose and apply appropriate techniques to graph a straight line given its equation
- Find the equation of a line that is parallel or perpendicular to a given line
- Solve linear inequalities and graph the solution on a number line

Gradient-Intercept Form and General Form

ⓘ Gradient-Intercept Form

Every linear relationship has this equation:

$$y = mx + c$$

coefficient of x / gradient — constant term / y-intercept

🔍 Determine gradient and y-intercept from an equation

$y = 2x + 7$	$y = 2x - 7$	$y = \frac{2x}{5} - 7$
Gradient: 2 y-intercept: $(0,7)$	Gradient: 2 y-intercept: $(0,-7)$	Gradient: $\frac{2}{5}$ y-intercept: $(0,-7)$

Determine the gradient and y-intercept of these lines.

a $y = 3x + 2$	**b** $y = 3x - 2$	**c** $y = -3x + 2$
d $y = 2 - 3x$	**e** $y = \frac{5x}{4} - 1$	**f** $y = 7x$

Determine equation of a line, given the gradient and y-intercept

Find the equation of a line with:

gradient 3 and y-intercept $(0,4)$.	gradient 3 and y-intercept $(0,-4)$.	gradient $-\frac{1}{2}$ and y-intercept $(0,0)$.
$y = 3x + 4$	$y = 3x - 4$	$y = -\frac{x}{2}$

Find the equation of a line with:

a	gradient 3 and y-intercept $(0,4)$.	**b**	Gradient is -2 and y-intercept is $(0,3)$	**c**	y-intercept is $(0,-1)$ and gradient is 5
d	y-intercept is $(0,2)$ and gradient is $\frac{2}{3}$	**e**	Gradient is 1 and y-intercept is $(0,-4)$	**f**	y-intercept is $(0,-9)$ and gradient is $-\frac{5}{6}$

Example Identify gradient and y-intercept after rearranging the equation

Rearrange the equation, then identify the gradient and y-intercept.

$y - 7 = 2x$

$y = 2x + 7$

Gradient: 2

y-intercept: 7

$3y + 15 - 2x = 0$

$3y = 2x - 15$

$y = \frac{2x}{3} - \frac{15}{3}$

Gradient: $\frac{2}{3}$

y-intercept: -5

Guided Practice

Rearrange the equation, then identify the gradient and y-intercept.

a $y + 2x = 5$

$m = -2, c = 5$

b $x - y = 7$

$m = 1, c = -7$

c $4x + 2y = 1$

$m = -2, c = 0.5$

d $-2x - 5y = 3$

$m = -\frac{2}{5}, c = -\frac{3}{5}$

General Form of a Line

The general form of a line is:

$$ax + by + c = 0$$

Where a, b, c are expressed as integers if possible, and a is preferred to be positive.

Identify General form of a line

✗	✓	✓	✗
$y = 3x + 5$	$3x - y + 5 = 0$	$99x + 25y = 0$	$-99x + 25y = 0$

Identify the lines written in general form:

a	$y = 2x + 3$	**b**	$2x - y + 3 = 0$	**c**	$x - 3y + 6 = 0$
d	$-x + 3y - 6 = 0$	**e**	$0.5x + y - 2 = 0$	**f**	$3x - 4y - \frac{1}{5} = 0$

Example Rearrange an equation of a line to general form.

Rearrange $y = 3x - 7$ to general form.

$-3x + y + 7 = 0$

$3x - y - 7 = 0$

Guided Practice

Rearrange to general form.

a	$y = 5x - 2$ $5x - y - 2 = 0$	**b**	$y = \frac{1}{2}x - 3$ $x - 2y - 6 = 0$
c	$y = 2x - \frac{3}{5}$ $10x - 5y - 3 = 0$	**d**	$y = \frac{x}{4} + \frac{1}{3}$ $3x - 12y + 4 = 0$

Checking if a Point Lies on a Line

A point lies on a line when its coordinates make the line's equation true.

1. Separate the **left-hand side LHS** and **right-hand side RHS** of the equation.
2. Substitute x and y values of the point into both sides.
3. Compare: If LHS = RHS, it lies on the line. If LHS ≠ RHS, it does not lie on the line.

Example Check whether a point lies on a line by substitution

Decide whether $(2,5)$ lies on the line

$$y = 2x + 1$$

$\text{LHS} = y$

$= 5$

$\text{RHS} = 2x + 1$

$= 2(2) + 1$

$= 5$

$\text{LHS} = \text{RHS}$

$(2,5)$ lies on the line $y = 2x + 1$

Decide whether $(-2,-5)$ lies on the line

$$y = 2x + 1$$

$\text{LHS} = y$

$= -5$

$\text{RHS} = 2x + 1$

$= 2(-2) + 1$

$= -3$

$\text{LHS} \neq \text{RHS}$

$(-2,-5)$ does not lie on line $y = 2x + 1$

Guided Practice

a Decide whether $(-2,4)$ lies on the line

$$y = -x + 2$$

$\text{LHS} =$

$\text{RHS} =$

yes

b Decide whether $(11,23)$ lies on the line

$$y = 35 - x$$

$\text{LHS} =$

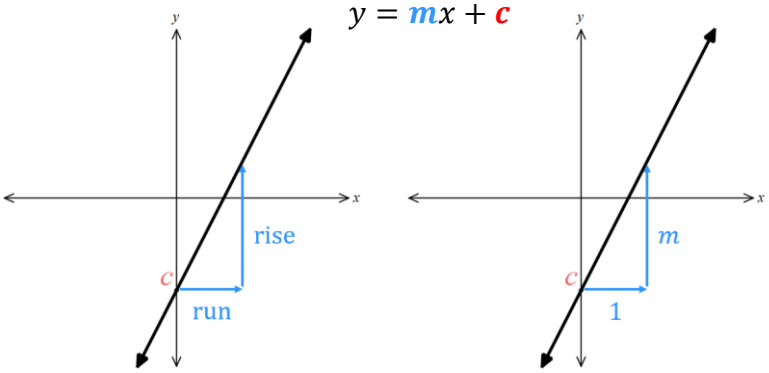
$\text{RHS} =$

no

Sketching Lines

ⓘ Sketching Lines without a Table

$y = mx + c$



1. Plot the y-intercept.
2. Starting from the y-intercept, use the gradient to find another point.
   a. The denominator is how much to move right (1 if the gradient is a whole number).
   b. The numerator is how much to move up or down.
3. Draw a straight line through both points and extend the line.

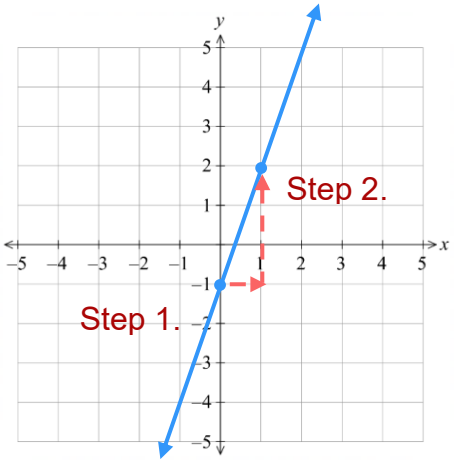
Example Sketch lines from the equation using gradient and y-intercept

Sketch $y = 3x - 1$

1. Plot a point at $(0, -1)$

2. The gradient is 3. Starting at $(0, -1)$:
   Move right 1 unit.
   Move up 3 units.

3. Join the two points and extend the line.

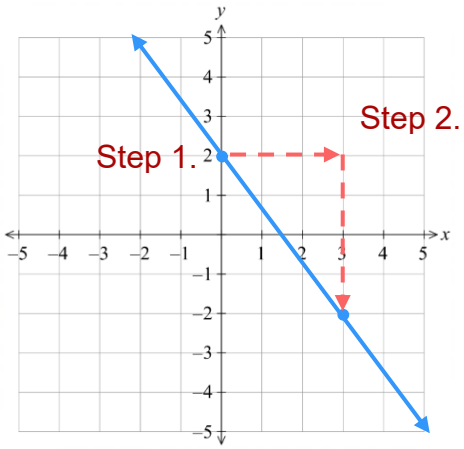


Sketch $y = -\frac{4}{3}x + 2$

1. Plot a point at $(0,2)$

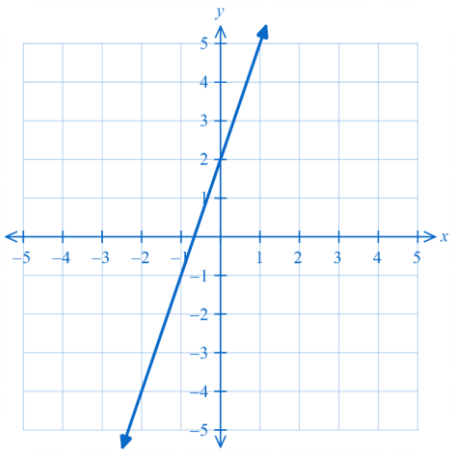
2. The gradient is $-\frac{4}{3}$. Starting at $(0,2)$:
   Move right 3 units.
   Move down 4 units.

3. Join the two points and extend the line.

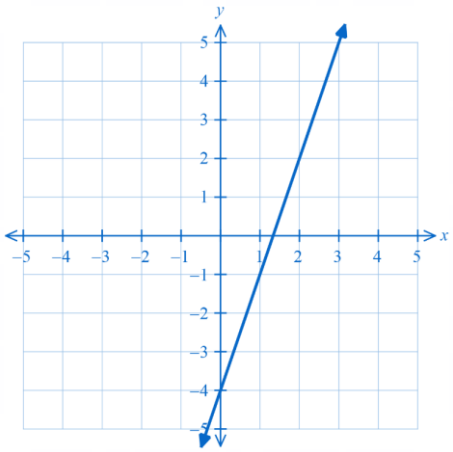


Guided Practice

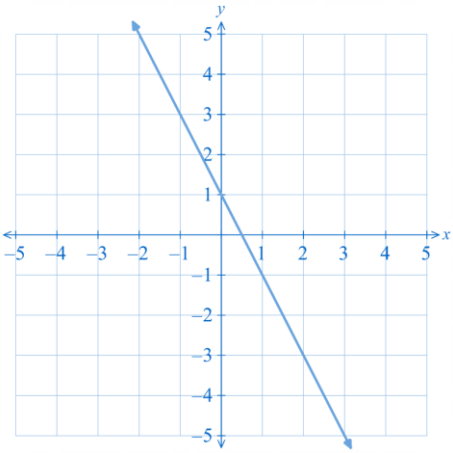
a Sketch $y = 3x + 2$



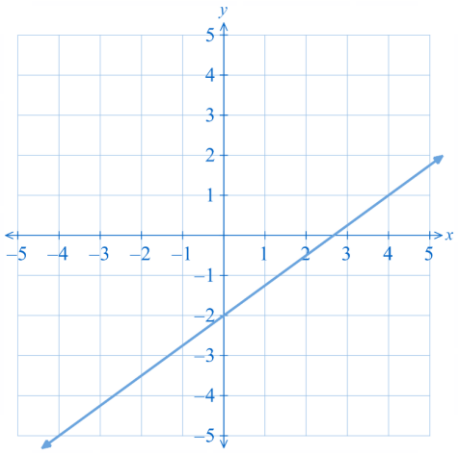
b Sketch $y = 3x - 4$



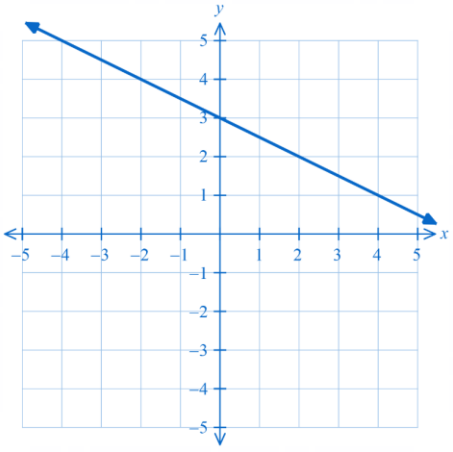
c Sketch $y = -2x + 1$



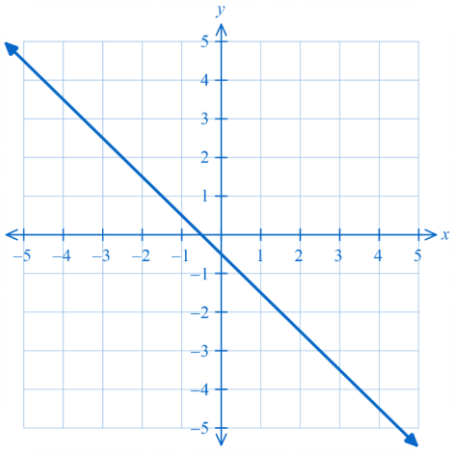
d Sketch $y = \frac{3x}{4} - 2$



e Sketch $y = -\frac{x}{2} + 3$



f Sketch $y = -x - \frac{1}{2}$

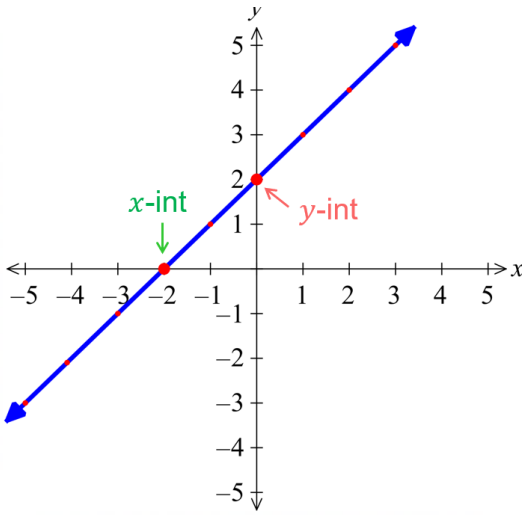


ⓘ x and y intercepts

x-intercept: where the line meets the x axis ($\boldsymbol{y = 0}$)

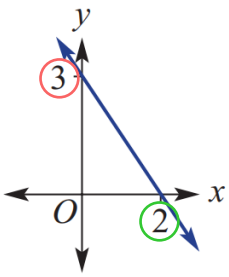
y-intercept: where the line meets the y axis ($\boldsymbol{x = 0}$)

		x-int					
$\boldsymbol{x}$	-3	-2	-1	0	1	2	3
$\boldsymbol{y}$	-1	0	1	2	3	4	5
				y-int			



Identify x and y intercepts of a line

a



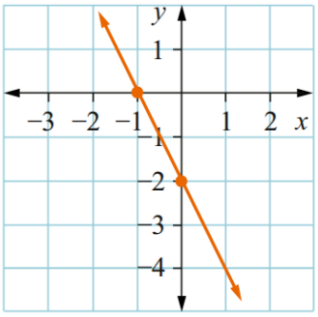
x-intercept: ..... y-intercept: .....

b

$\boldsymbol{x}$	-2	-1	0	1	2
$\boldsymbol{y}$	-6	-4	-2	0	2

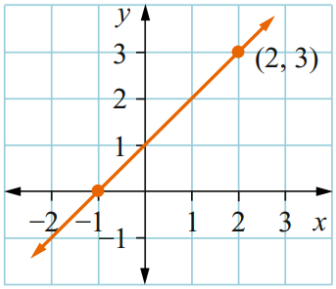
x-intercept: ..... y-intercept: .....

c



x-intercept: ..... y-intercept: .....

d



x-intercept: ..... y-intercept: .....

e

x	-2	-1	0	1	2
y	2	0	-2	-4	-6

x-intercept: ..... y-intercept: .....

f

x	-2	-1	0	1	2
y	-9	-6	-3	0	3

x-intercept: ..... y-intercept: .....

Finding the x and y intercepts

y intercept: substitute $x = 0$ and solve.

x intercept: substitute $y = 0$ and solve.

Example Identify x and y intercept from the equation

Find the x and y intercepts of

$$y = 4x + 2$$

y-int: $x = 0$

$y = 4(0) + 2$

$y = 2$

y-int is at $(0,2)$

x-int: $y = 0$

$0 = 4x + 2$

$-2 = 4x$

$-\frac{2}{4} = x$

$-0.5 = x$

x-int is at $(-0.5\ ,0)$

Find the x and y intercepts of

$$x + 2y - 6 = 0$$

y-int: $x = 0$

$0 + 2y - 6 = 0$

$2y = 6$

$y = 3$

y-int is at $(0,3)$

x-int: $y = 0$

$x + 2(0) - 6 = 0$

$x - 6 = 0$

$x = 6$

x-int is at $(6\ ,0)$

Guided Practice

Find the x and y intercepts of these lines.

a $y = 2x - 8$

y int: $x = 0$

$y = 2(0) - 8$

x int: $y = 0$

$0 = 2x - 8$

x-int: $(4,0)$, y-int: $(0,-8)$

b $x - y = 7$

y int: $x = 0$

x int: $y = 0$

x-int: $(7,0)$, y-int: $(0,-7)$

c $-3x - 2y = 6$

x-int: $(-2,0)$, y-int: $(0,-3)$

d $4x + 4y = 7$

x-int: $\left(\frac{7}{4},0\right)$, y-int: $\left(0,\frac{7}{4}\right)$

Sketching lines using the intercepts

1. Find the y intercept ($x = 0$)
2. Find the x intercept ($y = 0$)
3. Connect the x and y intercept and extend the straight line.

Example Sketch graphs from the equation by finding the intercepts

Sketch $x + 2y + 6 = 0$

1. y-int: $x = 0$

$2y + 6 = 0$

$2y = -6$

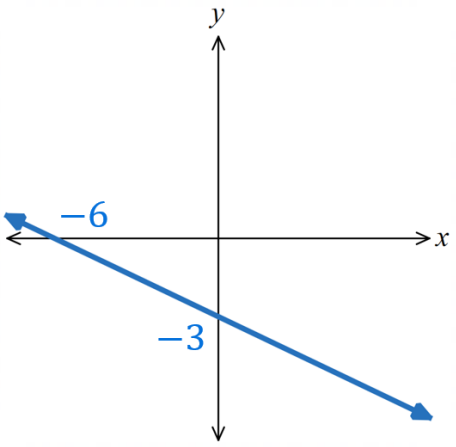
$y = -3$

2. x-int: $y = 0$

$x + 6 = 0$

$x = -6$

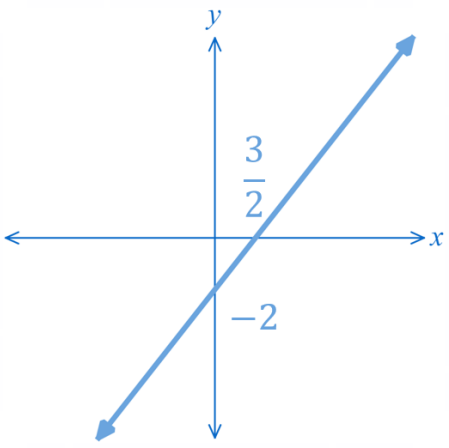
3.



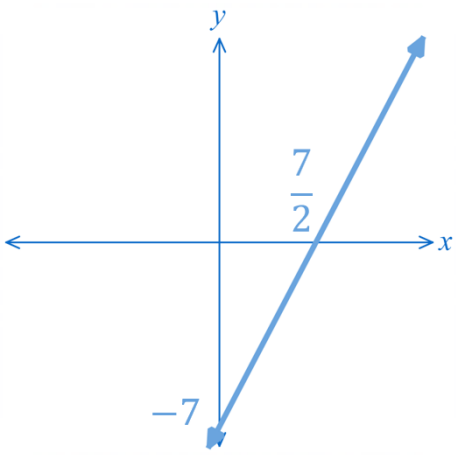
Guided Practice

Sketch by finding the intercepts:

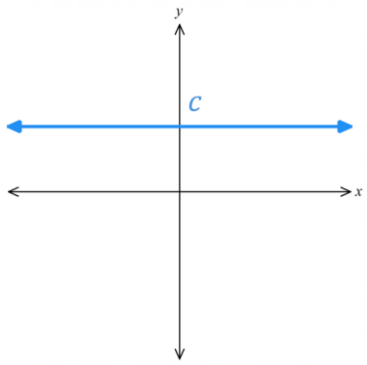
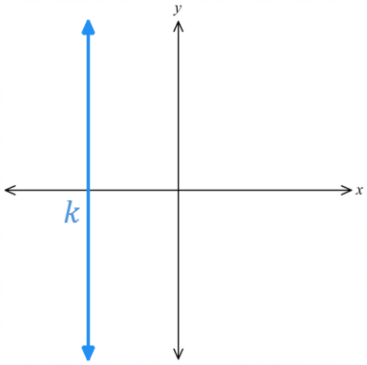
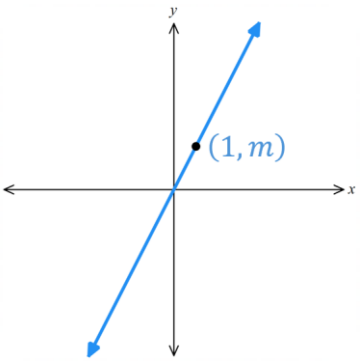
a $3y = 4x - 6$



b $4x - 2y = 14$



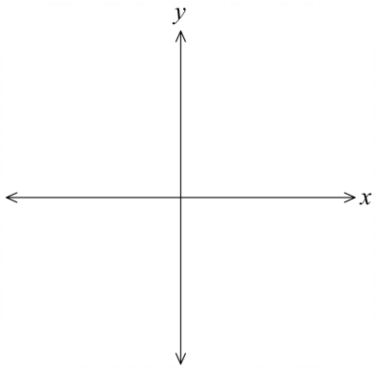
Special Lines

Horizontal Line	Vertical Line	Straight line through origin
$y = c$	$x = k$	$y = mx$
		

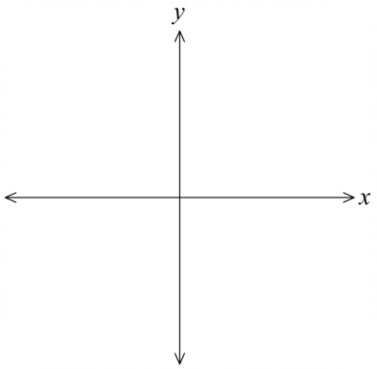
Sketch Special lines

For each equation, sketch the line, showing the intercept for horizontal/vertical lines, and the coordinate at $x = 1$ for lines through the origin.

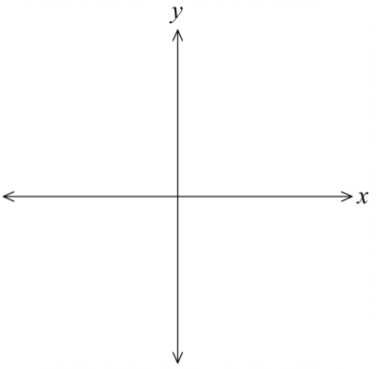
a $x = -1$



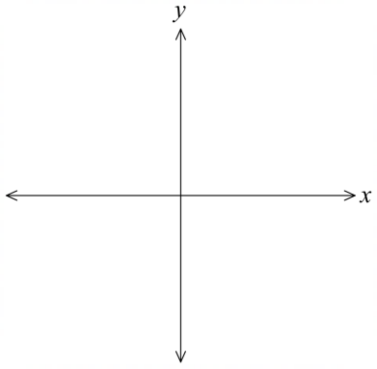
b $y - 2 = 0$



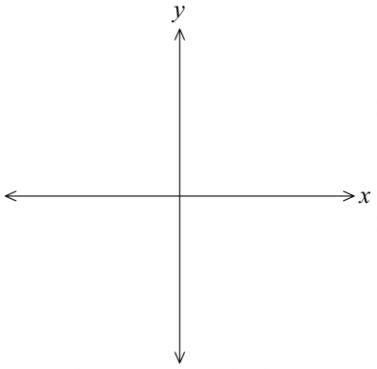
c $2y = -3$



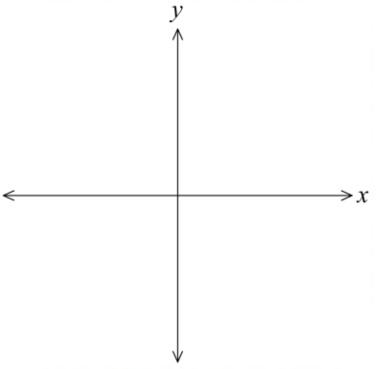
d $x + 2y = 0$



e $3x - 5y = 0$



f $2x = 5$



Point-Gradient Form

Point-Gradient Formula

The point-gradient formula of a line relates the gradient m and any point on the line (x_1, y_1)

gradient

$$y - y_1 = m(x - x_1)$$

y coordinate of a point

x coordinate of a point

Example Find equation of a line given gradient and a point

Find the equation of the line with gradient 3 passing through $(6,4)$.

m $x_1\ y_1$

$$y - y_1 = m(x - x_1)$$
$$y - 4 = 3(x - 6)$$
$$y - 4 = 3x - 18$$
$$y = 3x - 14$$

Find the equation of the line with gradient -4 and x intercept 1.

m $(x_1\ y_1) = (1,0)$

$$y - y_1 = m(x - x_1)$$
$$y - 0 = -4(x - 1)$$
$$y = -4x + 4$$

Guided Practice

Find the equation of the line, in gradient-intercept form, with:

a Gradient 4 and passes through $(-1,2)$

$y = 4x + 6$

b Gradient 3 and passes through $(-3,5)$

$y = 3x + 14$

c Gradient $\frac{1}{3}$ and passes through $(9,2)$

$y = \frac{x}{3} - 1$

d Gradient -3 passes through $(8,-5)$

$y = -3x + \frac{13}{3}$

Finding Equation of a Line given Two Points

1. Calculate gradient using the formula: $m = \frac{y_2 - y_1}{x_2 - x_1}$
2. Substitute into point-gradient form. $y - y_1 = m(x - x_1)$
3. Rearrange to $y = mx + c$.

Example Find equation of a line given two points

Find the equation of the line, in gradient-intercept form, passing through $(1, 5)$ and $(4, -1)$

1. $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-1-5}{4-1}$

$= -2$

2. Sub $m = -2$ and (1,5)

$y - y_1 = m(x - x_1)$

$y - 5 = -2(x - 1)$

3. $y - 5 = -2x + 2$

$y = -2x + 7$

Guided Practice

Find the equation of the line, in gradient-intercept form, passing through:

a (2,7) and $(-1.5, 0)$

1. $m = \frac{y_2 - y_1}{x_2 - x_1}$

2. $y - y_1 = m(x - x_1)$

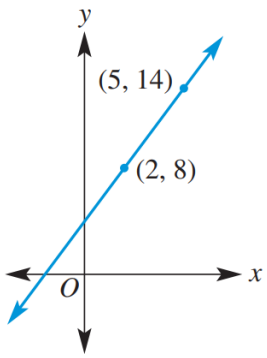
3.

$y = 2x + 3$

b $(1, 7)$ and $(3, -1)$

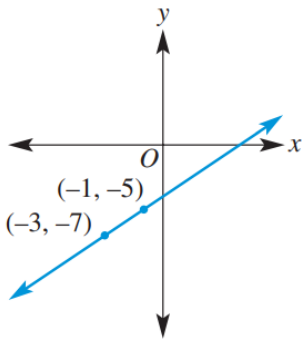
$y = -4x + 11$

c



$y = 2x + 4$

d



$y = x - 4$

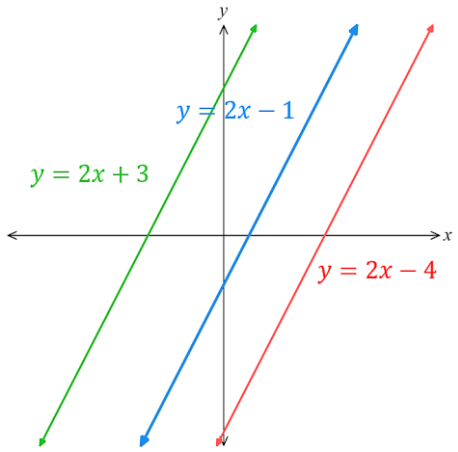
Parallel and Perpendicular Lines

ⓘ Parallel and Perpendicular Gradients

Parallel lines have the **same gradient**.
Perpendicular lines have gradients that **multiply to -1.**
Consider two lines $y = m_1x + c_1$ and $y = m_2x + c_2$.

Parallel Lines

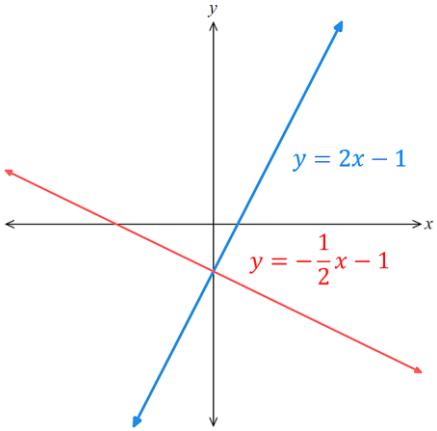
$$m_1 = m_2$$



Perpendicular Lines

$$m_1 \times m_2 = -1$$

$$m_1 = -\frac{1}{m_2}$$



🔍 **Identify** whether two lines are parallel, perpendicular, or neither

$y = 3x - 1$ $\quad m_1 = 3$ $y = 3x + 4$ $\quad m_2 = 3$ Parallel	$y = 4x - 2$ $\quad m_1 = 4$ $y = -\frac{1}{4}x + 3$ $\quad m_2 = -\frac{1}{4}$ Perpendicular	$y = 2x + 1$ $\quad m_1 = 2$ $y = -2x - 5$ $\quad m_2 = -2$ Neither

Decide whether these lines are parallel, perpendicular, or neither.

a	$y = -\frac{1}{5}x + 1$ $y = 5x + 2$	**b**	$y = 2x + 1$ $y = 2x - 100$	**c**	$y = 7x - 2$ $y = -7x + 2$
d	$y = 10 - 2x$ $y = 10 + 2x$	**e**	$y = \frac{3}{4}x$ $y = -\frac{4}{3}x$	**f**	$y = 3$ $x = 3$

Finding the Equation of a Parallel and Perpendicular Line

Gradient of a Parallel Line

$$m_{\text{par}} = m_{\text{original}}$$

Gradient of Perpendicular Line

$$m_{\text{perp}} = -\frac{1}{m_{\text{original}}}$$

1. Determine gradient m_{new} of the parallel/perpendicular line.
2. Write the equation $y - y_1 = m_{\text{new}}(x - x_1)$
3. Substitute new gradient and given point into the formula.
4. Rearrange to $y = mx + c$

Example Find the equation of a parallel/perpendicular line

Find the equation of a line, in gradient-intercept form, that is:

Parallel to $y = -2x - 7$ through $(1,9)$

1. $m_{\text{new}} = -2$
2. $y - y_1 = m_{\text{new}}(x - x_1)$
3. $y - 9 = -2(x - 1)$
4. $y - 9 = -2x + 2$

   $y = -2x + 11$

Perpendicular to $y = \frac{3}{4}x - 1$ through $(3, -2)$

1. $m_{\text{new}} = -\frac{4}{3}$
2. $y - y_1 = m_{\text{new}}(x - x_1)$
3. $y - (-2) = -\frac{4}{3}(x - 3)$
4. $y + 2 = -\frac{4x}{3} + 4$

   $y = -\frac{4x}{3} + 2$

Guided Practice

Find the equation of the line, in gradient-intercept form that is:

a Parallel to $y = x + 3$ and passes through $(1, 5)$

1. $m_{\text{new}} =$ .....
2. $y - y_1 = m_{\text{new}}(x - x_1)$
3.
4.

$y = x + 4$

b Parallel to $y = \frac{2}{3}x + 1$ and passes through $(3, -4)$

1. $m_{\text{new}} =$ .....
2. $y - y_1 = m_{\text{new}}(x - x_1)$
3.
4.

$y = \frac{2x}{3} - 6$

c Perpendicular to $y = 2x + 3$ and passes through $(2, 5)$

$y = -\frac{x}{2} + 6$

d Perpendicular to $y = -4x + 1$ and passes through $(-4, -3)$

$y = \frac{x}{4} - 2$